

Erik A. Roberts, MD, PhD

Clinical Assistant Professor

University of Michigan, Department of Anesthesiology

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Education and Training

Education

08/2008 – 06/2012

BS, University of California San Diego, San Diego, CA

08/2012 – 05/2021

MD-PhD, Boston University, Boston, MA

Postdoctoral Training

06/2021 – 07/2025

Anesthesiology Residency,

University of Michigan, Ann Arbor, MI

07/2025 – 07/2026

Cardiothoracic Anesthesiology Fellowship,

University of Michigan, Ann Arbor, MI

Certification and Licensure

Certification

03/2026 – 12/2031

Diplomate of the American Board of Anesthesiology

Licensure

02/2025 – 02/2028

MI, Physician License

02/2025 – 02/2028

MI, Controlled Substance License

02/2025 – 02/2028

DEA Controlled Substance Registration

Work Experience

Academic Appointment

07/2026 – Present

Clinical Assistant Professor,

University of Michigan, Ann Arbor, MI

Research Interests

1. Perioperative Outcomes
2. Perioperative Machine Learning

Clinical Interests

1. Cardiothoracic Anesthesiology
2. High-Risk Perioperative Care

Honors and Awards

National

2025

Best of Meeting Oral Abstract Presentation,

Society of Cardiovascular Anesthesiologists Annual Meeting,

Society of Cardiovascular Anesthesiologists

Institutional

2022	Michigan Medicine Making a Difference Award, University of Michigan
2023	Michigan Medicine Making a Difference Award, University of Michigan
2025	Michigan Medicine Making a Difference Award, University of Michigan

Departmental

2025	Peter J. Cohen Resident Research Award, University of Michigan Department of Anesthesiology
2025	Academic Resident of the Year Award, University of Michigan Department of Anesthesiology

Abstract Review

Conferences

2024	International Anesthesia Research Society Annual Meeting
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Teaching

Teaching Activity

Institutional

01/2016 – 06/2017	TA for Control Systems in Biomedical Engineering Course, Boston University
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Memberships in Professional Societies

2021 – 2026	Member, American Society of Anesthesiologists
2023 – 2026	Member, Society of Cardiovascular Anesthesiologists

Volunteer Service

2021 – Present	Mentor, American Society of Anesthesiology, Advised twelve medical students on anesthesiology residency applications and professional development.
2024 – 2025	Mentor, Operation Opportunity, Provided longitudinal mentorship to a first-generation, college-bound high school student pursuing a career in healthcare.

Scholarly Activities

Extramural Invited Presentations

1. **Roberts E. A.**, Mathis M. (04/2025) Latent Subtypes of Cardiac Surgery-Associated Chronic Kidney Disease Progression: A Multicenter Machine Learning Study of Granular Perioperative Features. Society of Cardiovascular Anesthesiologists, Montreal, Canada.

Intramural Presentations

1. Etomidate and Ketamine, **Roberts E. A.**, Overview of etomidate and ketamine pharmacology and clinical applications, University of Michigan Department of Anesthesiology Morning Conference, 04/2022, Ann Arbor, MI.

2. Local Anesthetic Systemic Toxicity, **Roberts E. A.**, Woodrum D., Reviewed pathophysiology and management of local anesthetic systemic toxicity, University of Michigan Department of Anesthesiology Morning Conference, 08/2022, Ann Arbor, MI.
3. Perioperative Risk Calculators, **Roberts E. A.**, Ragheb J., Introduced commonly used perioperative risk prediction tools and their clinical utility, University of Michigan Department of Anesthesiology Morning Conference, 11/2022, Ann Arbor, MI.
4. Research Journal Club: Sacral Anatomical Interspace Landmark for Lumbar Puncture in Pregnancy, **Roberts E. A.**, Pancaro C., Discussed literature on using anatomical landmarks for neuraxial anesthesia in obstetrics, University of Michigan Department of Anesthesiology Morning Conference, 02/2022, Ann Arbor, MI.
5. Non-OR Sedation, **Roberts E. A.**, Bukowski B., Reviewed sedation practices and safety considerations outside the operating room, University of Michigan Department of Anesthesiology Morning Conference, 01/2023, Ann Arbor, MI.
6. Dexmedetomidine and Spinal Anesthesia, **Roberts E. A.**, Kozlansky L., Reviewed evidence for using dexmedetomidine as an adjunct in spinal anesthesia, University of Michigan Department of Anesthesiology Morning Conference, 03/2023, Ann Arbor, MI.
7. Chronic Post Surgical Pain in Ambulatory Surgery, **Roberts E. A.**, Meral R., Journal club discussion of a study examining chronic postoperative pain after ambulatory surgery, University of Michigan Department of Anesthesiology Morning Conference, 08/2024, Ann Arbor, MI.
8. Call Me Maybe?: An After-hours Mott PACU Misadventure, Ballout N., Kramer Z., Pohl N., **Roberts E. A.**, Sleeper L., Root cause analysis of a safety event in the post-anesthesia care unit, University of Michigan Department of Anesthesiology Grand Rounds, 04/2025, Ann Arbor, MI.
9. Acute Perioperative Management of Maxillofacial and Neck Trauma, **Roberts E. A.**, Neevel A., Casper K., Morrison R., Khatibi N., Discussed a multidisciplinary approach to airway management in maxillofacial and neck trauma, University of Michigan Departments of Anesthesiology and Otolaryngology Combined Grand Rounds, 2025, Ann Arbor, MI.
10. Anesthetic Agents in Cardiac Surgery, **Roberts E. A.**, Korenke M., Review of anesthetic pharmacology and applications in cardiac surgery, University of Michigan Cardiac Anesthesiology Morning Conference, 10/2025, Ann Arbor, MI.
11. Perioperative CNS Dysfunction & Protection, **Roberts E. A.**, Fennema J., Overview of perioperative neurologic dysfunction in cardiac surgery, University of Michigan Cardiac Anesthesiology Morning Conference, 01/2026, Ann Arbor, MI.
12. Perioperative Spinal Cord Protection, **Roberts E. A.**, Wu B., Review of epidemiology and prevention of spinal cord ischemia in aortic surgery, University of Michigan Cardiac Anesthesiology Morning Conference, 03/2026, Ann Arbor, MI.

Publications/Scholarship

Peer Reviewed

Journal Articles

1. **Roberts, E. A.**, Troiano, C., & Spiegel, J. H. (2016). Standardization of Guidelines for Patient Photograph Deidentification. *Annals of plastic surgery*, 76(6), 611-614.

2. Kondabolu, K., **Roberts, E. A.**, Bucklin, M., McCarthy, M. M., Kopell, N., & Han, X. (2016). Striatal cholinergic interneurons generate beta and gamma oscillations in the corticostriatal circuit and produce motor deficits. *Proceedings of the National Academy of Sciences*, 113(22), E3159-E3168.
3. Howe, W. M., Gritton, H. J., Lusk, N. A., **Roberts, E. A.**, Hetrick, V. L., Berke, J. D., & Sarter, M. (2017). Acetylcholine release in prefrontal cortex promotes gamma oscillations and theta–gamma coupling during cue detection. *Journal of Neuroscience*, 37(12), 3215-3230.
4. Sherfey J. S., Soplata A. E., Ardid S., **Roberts E. A.**, Stanley D. A., Pittman-Polletta B. R. & Kopell N. J. (2018) DynaSim: A MATLAB Toolbox for Neural Modeling and Simulation. *Front. Neuroinform.* 12:10.

Chapters

1. Kondabolu, K., Kowalski, M. M., **Roberts, E. A.**, & Han, X. (2015). Optogenetics and deep brain stimulation neurotechnologies. In *Cognitive Enhancement* (pp. 441-450). Springer International Publishing.
2. Ramineni, A., **Roberts, E. A.**, Vora, M., Mahboobi, S. K., & Nozari, A. (2021). Anesthesia Considerations in Neurological Emergencies. *Neurologic Clinics*, 39(2), 319-332.

Non-Peer Reviewed

PhD Thesis

1. **Roberts, E. A.** (2019). *Global neural rhythm control by local neuromodulation* (Doctoral dissertation, Boston University).

Abstracts/Posters

1. Poster: **Roberts E. A.**, Salisbury K., Chan S., & Blevins N. H. (2011). Tissue Modeling in a Patient-Specific Skull Base Surgical Simulator. *Otolaryngology — Head and Neck Surgery*. San Francisco, CA.
2. Poster: **Roberts E. A.**, Kondabolu K., Abdulkarim M., McCarthy M. M., Kopell N., & Han X. (2015). Enhanced cortico-striatal beta oscillations and synchrony mediated by optogenetic increase of striatal cholinergic tone. 2015 Neuroscience Meeting Planner. Chicago, IL: Society for Neuroscience.
3. Poster: **Roberts E. A.**, Kopell N. J. (2017) GIMBL-Vis: A GUI-Based Interactive Multidimensional Visualization Toolbox for Matlab. *Organization for Computational Neurosciences*. Antwerp, Belgium.
4. Poster: **Roberts E. A.**, Mathis M. (04/2026) Association of Post-Cardiopulmonary Bypass Oxygen Delivery Index with Risk of Cardiac Surgery-Associated Acute Kidney Injury. Society of Cardiovascular Anesthesiologists, Nashville, TN.